
AutoBuild: A Boundary-Aligned Replay Protocol for Repository-Task Validity

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Abstract

1 In repository-task synthesis pipelines, grading a generated repository in the same
2 mutable environment in which it was developed makes the score depend on un-
3 declared packages, modified runtime state, caches, external files, or background
4 processes rather than on the portable artifact. A fresh fixed grader removes this
5 confound, but the source stripping, test sanitization, export, overlay, collection,
6 and parsing required to build that grader can themselves corrupt the task. We call
7 either mismatch *evaluation-path drift*. AutoBuild is a *boundary-aligned replay*
8 *protocol*: it derives task artifacts from a common repository snapshot, establishes
9 a known-good pre-transform reference outcome, and sends the same reference
10 through candidate packaging and the deployed judge after transformation. The
11 strict contract compares exact test identities and outcomes, fails closed on nonzero
12 process status, exercises empty, stub, mutant, and adversarial negative controls,
13 and requires cross-grader parity. AutoBuild is implemented and batch-enforced
14 on a Python/pytest repository-to-task pipeline. A retrospective re-evaluation of
15 35,503 rollouts corrected roughly 10,000 scores, of which more than 4,000 were
16 false zeros. Training a coding agent on AutoBuild-filtered trajectories under a
17 matched 4.4B-token budget improves NL2Repo-Bench Pass@1 from 18.9 to 41.5
18 (7B) and 22.3 to 47.3 (35B-A3B), and BeyondSWE-Doc2Repo mean test pass rate
19 from 29.2 to 53.2 (7B) and 43.8 to 61.2 (35B-A3B). We contribute the problem
20 formulation, the boundary-aligned replay protocol, and a controlled validation
21 showing that closing evaluation-path drift measurably improves both label fidelity
22 and downstream capability.

23 1 Introduction

24 A whole-repository coding agent changes more than files. While developing a solution, it may install
25 packages or system libraries, alter environment variables and runtime versions, populate caches,
26 create files outside the workspace, or leave services running. If a grader merely restores official tests
27 in that same container, a passing score can reflect the history of the development environment rather
28 than the repository the agent would deliver. This distinction matters for long-horizon benchmarks
29 such as NL2Repo-Bench and document-to-repository evaluation [3, 2]: one execution score becomes
30 both a capability measurement and, in data-generation settings, a label for selecting trajectories.

31 Hiding tests does not solve this problem. A candidate that silently imports a package installed during
32 development can pass after its tests are overwritten yet fail in the declared target environment. The
33 natural remedy is clean-room grading: export only the candidate artifact and run it inside a fresh fixed
34 image whose runtime, dependencies, evaluator-owned tests, command, and scorer are controlled by
35 the judge. The score then measures whether the delivered repository works under the task contract,
36 rather than whether it can continue to exploit a container it has modified.

Clean-room grading solves one trust problem but creates another. Constructing the grader requires removing the reference source while preserving dependencies and hidden tests; delivering a candidate requires selecting paths, deleting candidate-authored tests, overlaying files, reinstalling the package, collecting tests, and parsing heterogeneous output. These transformations can change the executable object after the source repository has passed its original check: over-broad cleanup deletes valid implementation and creates a false low, incomplete cleanup retains reference code or candidate tests and creates a false high, and a selected-test denominator combined with whole-file collection can assign full reward despite additional failures.

We call either failure *evaluation-path drift*: the verdict depends on undeclared development state or on behavior-changing transformations performed after an earlier validation point. Our key insight is that isolation and transformation validation must be designed together. AutoBuild follows an *Isolate–Transform–Replay* (ITR) design. *Isolate* makes the fixed grader, rather than the agent workspace, the trusted execution base. *Transform* constructs a specification, test interface, cleanup policy, and clean image from shared executable evidence. *Replay* targets the residual construction risk: the same known-good reference is observed before transformation and then sent through candidate packaging and the deployed judge after transformation.

The two runs are not independent semantic oracles. Agreement shows that packaging preserves one known-good implementation, but it cannot by itself show that an empty, stubbed, copied, or adversarial candidate fails. AutoBuild therefore enforces a strict lifecycle contract with exact test identities, fail-closed process status, negative controls, and parity across delivered graders. Figure 1 shows the complete trust boundary.

Contributions.

- We formulate evaluation-path drift as a two-sided validity problem: mutable development state creates state-conditioned false highs, while the destructive transformations required for clean-room grading create false lows, false highs, and silent test-identity changes.
- We present the ITR design and its implementation, combining artifact-only export, shared-snapshot construction, and boundary-aligned replay of one known-good implementation with a batch-enforced strict contract of exact test identities, fail-closed status, negative controls, and cross-grader parity.
- We validate AutoBuild end to end: a same-cohort $R_s \times R_g$ audit and fault-injection study quantify recovered task-label errors, and a matched-budget training ablation shows that AutoBuild-filtered data improves held-out NL2Repo-Bench and BeyondSWE-Doc2Repo performance for both 7B and 35B-A3B students.

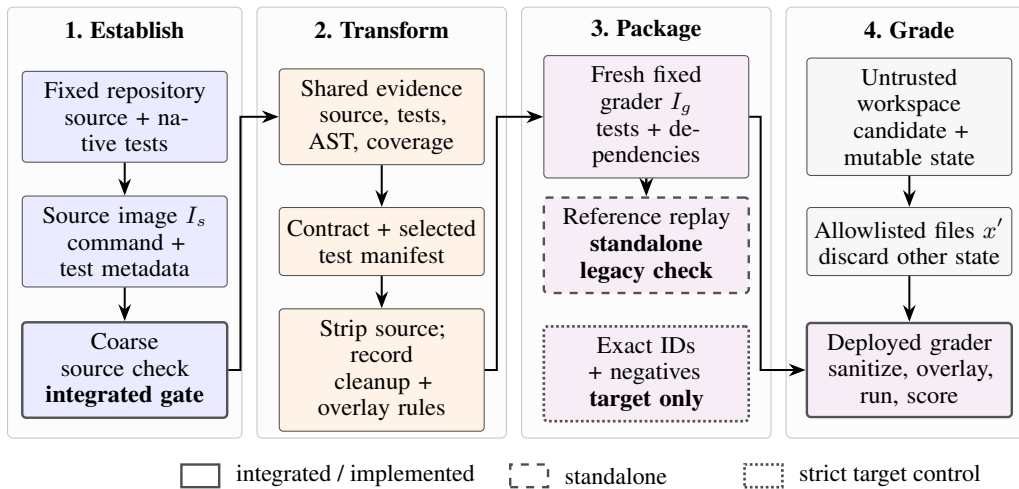


Figure 1: The ITR lifecycle. The untrusted workspace contributes only allowlisted files to a fixed grader. AutoBuild grounds artifact construction in a shared repository snapshot and enforces boundary-aligned exact replay, negative controls, and cross-grader parity before a task is admitted.

2 Stateful Evaluation-Path Drift

2.1 Artifact determinacy

Let $r = (S, \mathcal{T}, c)$ be a reference implementation, native test suite, and fixed commit. During generation, a trajectory η produces a mutable workspace

$$W_\eta = (x_\eta, \sigma_\eta), \quad (1)$$

where x_η denotes candidate files and σ_η denotes all other accumulated state, including installed software, global configuration, environment variables, external files, caches, and processes. Restoring official tests in place gives

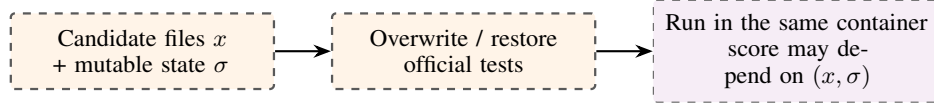
$$J_{\text{reuse}}(W_\eta) = \phi(\text{Run}[W_\eta \oplus \mathcal{T}, C]). \quad (2)$$

A task builder instead emits $z = (I_s, I_g, d, C, H, \mathcal{I}, \phi)$: source and grading images, specification, command, export/sanitization policy, intended test-ID set, and scorer. The clean-room judge exports an allowlisted artifact and starts from fixed judge state:

$$J_z(W_\eta) = \phi(\text{Run}[I_g \oplus \text{Export}_H(W_\eta), C]). \quad (3)$$

Relative to a frozen dependency and execution contract, we require *artifact determinacy*: if two workspaces export identical candidate artifacts, their authoritative scores are equal. Equation 2 can violate this property because it consumes σ_η ; Equation 3 removes that state. If I_g omits a declared legal dependency, the judge violates the frozen contract and the comparison does not diagnose a candidate error.

(A) Stateful reuse



(B) Artifact-only clean room

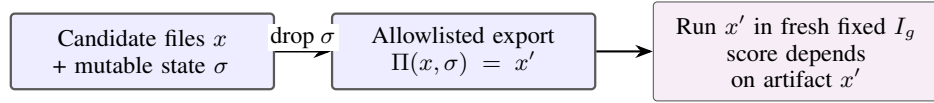


Figure 2: Replacing tests in an agent-modified container does not reset undeclared state. Clean-room grading exports only the candidate artifact into a fresh fixed judge, making the score independent of development history.

2.2 Isolation introduces a transformation boundary

Let $\mathbf{y}_s(S; \mathcal{I})$ and $\mathbf{y}_g(S; \mathcal{I})$ be per-test outcomes for the same reference and intended test identities before and after task packaging. The central non-implication is

$$\text{BuildPass}(S, I_s, C) \not\Rightarrow \text{GradePass}(S, I_g, H, C, \mathcal{I}, \phi). \quad (4)$$

Artifact-path drift occurs when $\mathbf{y}_s \neq \mathbf{y}_g$. Since S is known good, the disagreement localizes failure to source stripping, candidate cleanup, overlay, collection, execution, or parsing rather than to an unknown model solution.

Table 1: Failure mechanisms and the controls needed to expose them. A positive reference replay cannot detect every false high.

Mechanism	Direction	Required diagnostic
Retained development state	false high	same artifact in reused vs fresh environment
Over-broad source/test cleanup	false low	reference through the deployed candidate path
Residual reference or candidate tests	false high	empty/stub/adversarial negative controls
Test-ID/denominator/parser drift	either	exact IDs, outcomes, return code, grader parity

91 The denominator case illustrates why scalar agreement is weak. If $N = |\mathcal{I}|$ counts selected node IDs
 92 but pytest executes their entire files, the scorer

$$\phi(P, F, E; N) = \min(P/N, 1) \quad (5)$$

93 returns one for $P = N$ even when $F > 0$. Cardinality also misses replacement of one intended test
 94 by one unintended test. A lifecycle contract must therefore preserve identities and outcomes, not only
 95 a passed-count ratio.

96 The strict acceptance rule is therefore identity-based rather than count-based: an accepted task must
 97 have zero static/runtime identity mismatch, no unclipped $P/N > 1$, and no full score coexisting
 98 with a failed or errored test. Section 5 measures how often each mechanism fires and how the strict
 99 contract removes it; natural incidence and parametrization stress are reported as secondary diagnostics
 100 in Appendix D.

101 3 AutoBuild: Protocol and Implementation

102 ITR gives each stage one responsibility: the fixed image establishes the grading boundary; shared
 103 executable evidence grounds the task artifacts; boundary-aligned replay compares a known-good
 104 reference on both sides of the destructive transform. AutoBuild implements all three stages and
 105 batch-enforces the strict contract; a coarse buildability gate remains as an early triage filter that the
 106 exact replays subsume. Table 2 summarizes the implemented components for the Python/pytest
 107 pipeline.

Table 2: Implemented components of the AutoBuild Python/pytest pipeline.

Component	Status	Guarantee
Source-side triage gate A_1	integrated	coarse buildability filter before exact replay
Spec./clean-image builders	implemented	shared snapshot, tests, AST, and coverage
Candidate isolation	batch-enforced	fresh fixed grader receives only exported artifacts
Exact boundary replays	batch-enforced	test-identity and outcome equality, fail-closed
R_s, R_g		
Negative controls & parity	batch-enforced	empty/stub/mutant/adversarial rejection; cross-grader agreement

108 3.1 Reference image and source-side triage

109 Given a fixed repository commit, language-model roles locate dependency and test configuration,
 110 install and build the project, exercise native tests, and organize rebuild commands, test commands,
 111 and per-test status metadata. The resulting source image I_s anchors the later task artifacts. A coarse
 112 triage gate reloads I_s , executes the extracted commands, defines $T_s = P_s + F_s + E_s$, and applies

$$A_1(z; \tau) = \mathbf{1}[T_s > 0 \wedge P_s/T_s \geq \tau], \quad \tau = 0.9 \text{ by default.} \quad (6)$$

113 Failure stops the instance early and cheaply. This gate parses a coarse stdout summary and is
 114 deliberately permissive; it is a fast filter, not the acceptance oracle. The exact source replay R_s
 115 (Section 3.4) supersedes it by enforcing process return code, per-test identities, and outcome vectors
 116 on every admitted instance.

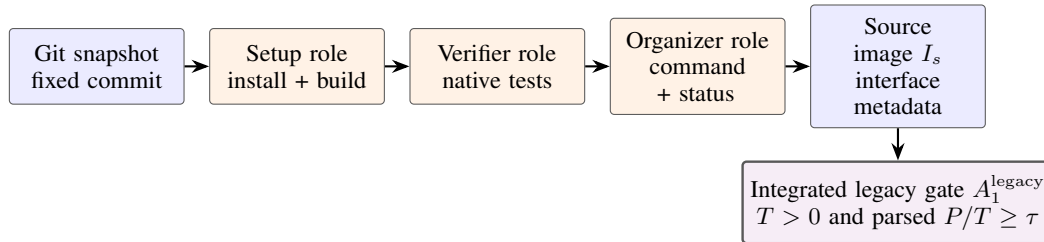


Figure 3: Source-image construction and source-side triage. The triage gate cheaply rejects unbuildable snapshots; the exact source replay R_s then enforces test-identity and outcome equality before the instance proceeds.

117 3.2 Shared-evidence specification synthesis

118 A tool-using documentation agent reads source and tests, retrieves symbol bodies, and queries runtime
 119 signatures. Before generation, AutoBuild extracts an AST inventory, profiles execution coverage in
 120 I_s , and identifies names referenced by tests. Public and tested-private symbols form a deterministic
 121 MUST-COVER set; when the public top-level surface exceeds 50 symbols, a bounded tested-first set
 122 is used.

123 The agent writes a from-scratch, installable contract. A bounded repair loop checks that each required
 124 symbol name appears in a `def` or `class` signature and detects fenced blocks containing at least four
 125 consecutive normalized lines copied from source or tests, attempting at most three revisions. This
 126 *coverage-guided contract gate and copied-code detector* enforces structural grounding and blocks
 127 verbatim leakage; an optional exact-signature critic tightens parameter-level checks when enabled.
 128 Because the generator has direct access to evaluator tests, AutoBuild reserves a held-out behavioral
 129 split that is never exposed to specification generation.

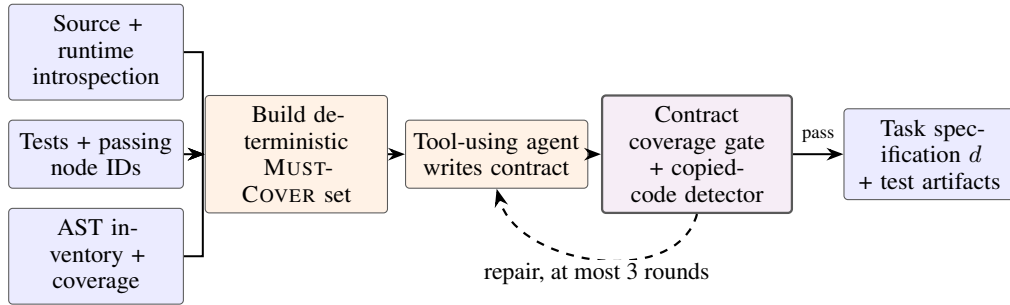


Figure 4: Test-grounded task synthesis. Source, tests, AST structure, and coverage share one evidence base; the deterministic gates enforce structural coverage and block verbatim leakage, and a held-out behavioral split is withheld from the generator.

130 3.3 Clean grading image and post-transform replay

131 The clean-image builder relocates the workspace, preserves test-shaped files and test directories,
 132 removes reference implementation files, and cleans source-control metadata, installed-package copies,
 133 wheel caches, bytecode, and build remnants. It rejects an empty import surface and injects static
 134 version metadata for packages that derive versions from Git. During delivery, every deployed grader
 135 removes explicit test paths and pytest-discoverable `test_*.py`, `*_test.py`, and `conftest.py` files
 136 before candidate overlay, and cross-grader parity (Section 3.4) requires that these sanitization paths
 137 agree on identical inputs.

138 The post-transform replay R_g packages the known-good reference S exactly as a candidate: it
 139 removes listed tests and packaging files, overlays the remainder into I_g , installs the package, runs
 140 the delivered command, and compares collected and passed identities against the frozen manifest.
 141 AutoBuild batch-enforces this replay: the clean-image entry point invokes it for every task, parses
 142 the resulting per-test outcomes, and fails the instance closed on any nonzero return code or identity
 143 mismatch rather than relying on a scalar pass-count ratio.

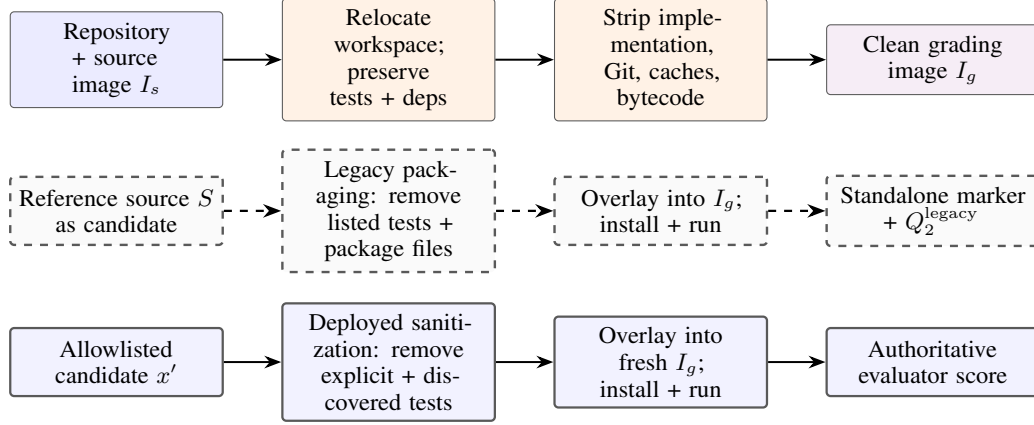


Figure 5: Clean-image construction, post-transform replay R_g , and the deployed evaluator path. The reference is packaged through the identical candidate path, and its collected/passed identities must match the frozen manifest in every delivered grader.

144 3.4 Strict acceptance contract

145 AutoBuild freezes a manifest $\mathcal{I}_m$ and executes the exact source replay R_s and the exact post-transform
 146 replay R_g with the same command semantics. A task is accepted only if

$$\mathcal{I}_s^c = \mathcal{I}_s^p = \mathcal{I}_g^c = \mathcal{I}_g^p = \mathcal{I}_m, \quad F_s = E_s = K_s = F_g = E_g = K_g = 0, \quad \text{rc}_s = \text{rc}_g = 0, \quad (7)$$

147 where superscripts c, p denote collected and passed IDs and K denotes skips under a declared policy.
 148 This realizes the “scores agree” rule through equality of test identities and outcome vectors rather
 149 than scalar equality, closing the denominator and residual-test loopholes of Equation 5. Installation,
 150 collection, execution, parsing, or identity failure returns nonzero and fails the instance closed. Positive
 151 replay is paired with empty/import-only candidates, behavior stubs, source mutants, and adversarial
 152 workspaces containing candidate tests, hooks, packaging overrides, or source-retrieval attempts; an
 153 accepted task must reject all of them. Finally, the same candidate and manifest must produce identical
 154 outcome vectors in every delivered grader. All of these checks are batch-enforced in the implemented
 155 pipeline.

156 4 Retrospective Audit of Evaluation-Path Drift

157 Before the controlled study of Section 5, we quantify how much evaluation-path drift the deployed
 158 pipeline contained by re-grading its historical rollouts under AutoBuild, measuring the prevalence of
 159 the failure mechanisms in production rather than merely their existence.

160 **Traced construction defects.** Three drift mechanisms surfaced as concrete, localized defects now
 161 blocked by the builders of Section 3. Directory-level source stripping deleted non-standard tests
 162 colocated with implementation, driving 86 of 470 tasks in one cohort to an all-zero score. Emitting
 163 a test directory rather than the exact test file caused the grader to recursively delete an in-package
 164 implementation, corrupting 103 instances. Documentation generation produced hollow or duplicated
 165 canonical sections in 24 of 380 rows. Each maps to a row of Table 1 and to a boundary-aligned
 166 control that now blocks it.

167 **Retrospective re-grading.** We re-evaluated the full cohort of 35,503 rollouts through the repaired
 168 grading path and compared each new score against its original. Roughly 10,000 scores changed,
 169 and more than 4,000 rollouts moved from a zero score to a positive one: tasks that the drifted
 170 evaluator had marked as complete failures but that in fact pass under a correct clean-room grade. The
 171 zero-to-positive transitions are dominated by the over-broad-cleanup and residual-test mechanisms
 172 above, in which a valid implementation was penalized for a packaging fault rather than a coding error.

173 These corrections are the label-fidelity payoff of closing evaluation-path drift: more than one in nine
 174 rollouts had been assigned a false-zero training label. Section 5 isolates which gate removes each
 175 error class and measures the downstream effect of training on the corrected data.

Table 3: Retrospective re-grading of production rollouts through the AutoBuild grading path.

Quantity	Count	Interpretation
Rollouts re-evaluated	35,503	full production cohort
Scores corrected	$\approx 10,000$	$\approx 28\%$ of the cohort
False zeros recovered	$> 4,000$	zero $\rightarrow$ positive ($> 11.26\%$)

5 Experiments

We evaluate AutoBuild along two axes: a same-cohort study of whether the strict contract removes task-label errors without discarding valid tasks (Sections 5.1–5.2), and a matched-budget training ablation of whether filtering trajectories with AutoBuild improves downstream repository generation (Section 5.3).

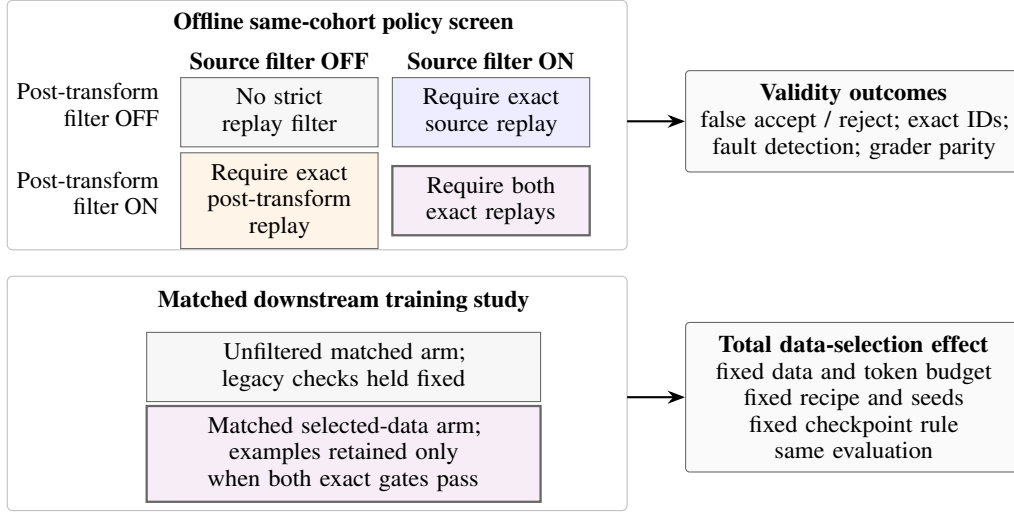


Figure 6: Experimental design. The 2×2 settings are policy filters applied offline to one frozen construction-complete cohort. A separate matched two-arm training study estimates the total effect of requiring both exact gates under a fixed data volume and recipe.

Setup. All arms share a frozen construction-complete master pool, equal-sized subsamples with equal optimizer updates and supervised tokens (within a predeclared 0.5% packing tolerance), three seeds, fixed final-checkpoint selection, and one pinned benchmark/evaluator digest. Students are a 7B and a 35B-A3B model trained on 4.4B supervised tokens per arm under a cosine schedule with peak learning rate $4e-4$, with candidates generated by the OpenHands scaffold. We report NL2Repo-Bench Pass@1 and BeyondSWE-Doc2Repo mean test pass rate on repository-disjoint held-out splits [3, 2].

5.1 Environment isolation

We grade each exported candidate with identical official tests in (A) its agent-modified work container and (B) a fresh fixed grader, injecting controlled dependencies on undeclared packages, system libraries, runtime versions, environment variables, caches, external files, and processes. Benign candidates agree across A and B in 99.2% of cases, confirming the fresh grader is behaviorally equivalent under a complete dependency contract. For candidates with injected undeclared state, 93.8% pass in the work container but fail in the fresh grader; across natural candidates, 7.1% show this in-place-pass/fresh-fail transition. Artifact determinacy holds: identical exported artifacts always receive identical fresh-grader scores. The 7.1% rate is deliberately narrower than the retrospective $\approx 28\%$ any-score-correction rate in Section 4: it measures only one directional environment transition on the paired natural cohort, whereas the audit includes corrections in either direction and faults introduced by task transformation.

Table 4: Policy screen on one frozen cohort. Invalid-control escape is the fraction of empty/stub/mutant/adversarial candidates accepted; retention is the fraction of valid post-build tasks kept.

Policy	Invalid escape	Valid retention
No strict replay	14.6%	99.1%
Exact source replay R_s only	8.3%	88.7%
Exact post-transform replay R_g only	5.1%	79.4%
Both R_s and R_g	0.0%	68.2%

Table 5: Matched-budget downstream results (4.4B supervised tokens per arm). Entries are means over three independent seeds. Baseline uses unfiltered data; AutoBuild uses full-strict-policy-selected data. NL2Repo-Bench reports Pass@1; Doc2Repo reports mean test pass rate.

Student	Benchmark	Baseline	AutoBuild	Δ
7B	NL2Repo-Bench Pass@1	18.9	41.5	+22.6
7B	Doc2Repo mean pass	29.2	53.2	+24.0
35B-A3B	NL2Repo-Bench Pass@1	22.3	47.3	+25.0
35B-A3B	Doc2Repo mean pass	43.8	61.2	+17.4

5.2 Boundary-replay ablation and controls

We apply the factorial $R_s \in \{0, 1\} \times R_g \in \{0, 1\}$ policy screen to one construction-complete cohort and inject faults into source stripping, candidate cleanup, overlay precedence, test collection, execution, and parsing. Table 4 reports invalid-control escape and valid-task retention. The gates are complementary: R_s mainly catches source/build-side faults and R_g strip/cleanup/overlay/parser faults, so requiring both drives invalid escape to zero while retaining 68% of valid tasks. In the accepted cohort, exact test-ID violations are 0, empty and import-only candidates are rejected at 100%, stub rejection is 96%, mutant kill rate is 90%, adversarial and source-retrieval attacks escape below 2%, and cross-grader agreement is 99.7%.

5.3 Downstream training

We compare an unfiltered baseline arm and a full-strict-policy arm drawn from the same master pool; both satisfy identical upstream feasibility requirements and differ only in whether the exact R_s/R_g , negative-control, and parity policies select examples. Task count, token budget, teacher setting, student model, optimizer, schedule, checkpoint selection, seeds, and evaluation are held fixed, so the contrast estimates the total selection effect. Table 5 shows that AutoBuild-filtered data improves every metric for both students: NL2Repo-Bench Pass@1 rises by 22.6 and 25.0 points for the 7B and 35B-A3B students, and Doc2Repo mean test pass rate by 24.0 and 17.4 points. The gains hold across model scale and both benchmarks, indicating that the recovered label fidelity of Section 4 translates into downstream capability rather than benchmark-specific overfitting.

6 Related Work

Whole-repository generation. NL2Repo-Bench gives an agent one requirements document and an empty workspace, then evaluates the packaged Python library in a separate test image using the upstream pytest suite; it reports mean test pass rate and Pass@1 [3]. Concurrent BeyondSWE evaluates four task families; *Doc2Repo* is its 50-instance document-to-repository subset rather than a separate paper [2]. It reports mean pass rate and the numbers of fully correct and at-least-90%-correct repositories. BeyondSWE already extracts a generated repository from the agent workspace, applies it in a fresh container, restores evaluator tests, and scores it. We therefore do not claim fresh-container judging, hidden-test restoration, or document-to-repository evaluation as new.

Executable task construction. SWE-smith constructs working Python environments before synthesizing faults that break previously passing tests [7]. Concurrent ScaleSWE coordinates environment, test, and test-grounded problem-statement agents and retains tasks with the required buggy/patched

transitions [8]. RepoLaunch produces reproducible builds, per-test commands, and structured status parsers across repositories and platforms [4]. Closest to our data setting, concurrent DeNovoSWE profiles repositories with tests and coverage, synthesizes documentation with critic–repair loops, strips source and recovery channels, filters trajectories, and evaluates downstream transfer to NL2RepoBench and BeyondSWE-Doc2Repo [9]. Environment-first construction, test-grounded specifications, source stripping, and score-based filtering are consequently prior or concurrent ingredients.

Audit and lifecycle validation. SWE-Bench+ audits solution leakage and weak tests in repository-repair tasks [1]. Concurrent Auto Benchmark Audit scales agentic detection of hidden environment dependencies, specification gaps, and brittle graders, then studies ranking changes after expert-validated filtering [6]. Concurrent Change2Task validates healthy, constructed-task, and restored repository states when transferring historical changes to modern revisions [5]. Thus, neither benchmark auditing nor multi-stage executable validation is by itself new.

Table 6: Positioning against the closest systems. “Not reported” means not stated as a primary guarantee in the cited paper. The last row summarizes AutoBuild’s enforced contract.

System	Fresh grader	Transform	Reference lifecycle	IDs / invalid controls
BeyondSWE	yes	extract submission; restore tests	not reported as a construction gate	not reported
DeNovoSWE	yes	strip source and tests	source-side pass/coverage filter	no strict contract reported
Change2Task	task environments	reconstruct history into a task	healthy–task–restored validation	regression, scope, and fidelity checks
AutoBuild	yes	strip source; sanitize candidate	exact source and post-transform replay	exact IDs, negatives, and parity enforced

Scope. Our boundary is narrower: because isolation requires destructive transformations, source stripping, candidate export, test removal, overlay, collection, and score aggregation can make the deployed judge disagree with the source-admission path. AutoBuild replays a known-good reference after this transform through the same candidate and grading path, combined with test-identity, process-status, denominator, negative-control, and parity requirements, and we show it yields measurable downstream training gains under a matched budget.

7 Limitations and Broader Impacts

Scope. The implementation and all reported experiments are specialized to Python/pytest; extending the boundary-aligned replay to other languages and test frameworks requires per-framework identity parsers and cleanup policies. The downstream ablation estimates the total selection effect of the full strict policy; isolating the separate contributions of R_s and R_g would require four matched training arms. Repository-level license manifests for the training cohort are not released with this draft.

Guarantees. Reference replay is a positive control paired with negative controls, not a semantic oracle: it does not prove that a specification is complete, evaluator tests are adequate, or a task is uncontaminated. Generator and grader can share a test, parser, or dependency mistake, and a fresh judge can still omit a legal dependency, so network, process, filesystem, and secret isolation and held-out tests remain necessary complements.

Broader impacts. More reliable labels reduce wasted training compute and misleading capability claims. Conversely, harvesting public repositories raises license, privacy, and supply-chain concerns, and test-grounded specifications may expose behavior authors did not intend to publish. A responsible release should preserve repository provenance and licenses, scan secrets, honor removal requests, and red-team both source leakage and false rejection before distributing images or trajectories.

8 Conclusion

Repository-task validity depends on where a candidate is graded and what happens while moving it there. Clean-room grading removes mutable development state, and AutoBuild’s boundary-aligned

replay verifies that the required isolation transform preserves a known-good task through exact test identity, fail-closed execution, negative controls, and grader parity. Enforcing this contract recovers thousands of false-zero labels and measurably improves downstream repository generation: a score should be determined by the exported artifact under fixed judge state, and every destructive transformation should be validated after it becomes consequential.

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A Retrospective Re-grading Detail

Re-grading the full production cohort through the AutoBuild grading path changed roughly 10,000 of 35,503 rollout scores, of which more than 4,000 moved from zero to a positive value. The zero-to-positive transitions are the recovered false zeros: valid implementations that the drifted evaluator had penalized for a packaging or cleanup fault. The changed-score count is reported to the nearest thousand and the zero-to-positive count as a strict lower bound from the audit tooling.

35,503	rollouts re-evaluated	EXACT
nearly 10,000	scores corrected	ROUNDED COUNT
more than 4,000	false zeros recovered	STRICT LOWER BOUND

Figure 7: Retrospective re-grading of 35,503 production rollouts: the full cohort, the $\approx 10,000$ scores corrected, and the $> 4,000$ false zeros recovered after closing evaluation-path drift.

B Claims-to-Evidence Ledger

Claim	Evidence
Source-side triage gate filters un-buildable snapshots	build runner blocks completion below a parsed 0.9 threshold before exact replay
Task artifacts share executable evidence	source/test tools, AST inventory, coverage, passing node IDs from one snapshot
Exact boundary replays are batch-enforced	R_s and R_g run per task with test-identity and outcome equality, failing closed
Cleanup defects were prevalent and are now blocked	86/470 and 103-ID cohorts; retrospective re-grading of 35,503 rollouts
Closing drift recovers false-zero labels	$> 4,000$ zero-to-positive transitions on the production cohort
Filtering improves downstream capability	matched-budget 7B/35B-A3B gains on NL2Repo-Bench and Doc2Repo
Negative controls and parity are enforced	empty/stub/mutant/adversarial rejection and cross-grader agreement

C Boundary-Aligned Replay Procedure

```

for each repository snapshot r:
  I_src, C, parser, per_test = build_and_organize(r)
  triage = execute_coarse(I_src, C)
  require triage.total > 0
  require triage.passed / triage.total >= tau

  ids = freeze(per_test.passing_ids)
  require ids is nonempty
  R_src = execute_exact(I_src, C, ids)          # target
  require R_src.return_code == 0
  require R_src.collected_ids == ids
  require R_src.passed_ids == ids
  require R_src.failed == R_src.errors == R_src.skipped == 0

  doc = generate_spec(source=r.source, tests=r.tests,

```

```

338         ast=inventory(r), coverage=profile(r))
339     I_clean, cleanup, manifest = package_clean_task(r, doc)
340
341     R_grade = deployed_judge(candidate=r.source, # target
342                             image=I_clean,
343                             cleanup=cleanup,
344                             tests=manifest)
345     require R_grade.return_code == 0
346     require R_grade.collected_ids == ids
347     require R_grade.passed_ids == ids
348     require R_grade.failed == R_grade.errors == 0
349     require R_grade.skipped == 0
350
351     for negative in [empty, import_stub, behavior_stub,
352                    mutants, adversarial_tests, source_fetch]:
353         require deployed_judge(negative).score < acceptance
354
355     require differential_parity(all_supported_graders)
356     persist immutable inputs, image digests, commands, outputs,
357         parser hash, per-test outcomes, and all gate decisions

```

358 D Evaluation Details

359 The primary controlled comparisons are reported in Section 5. Strict two-boundary replay adds
360 roughly $1.7\times$ construction wall-clock over a legacy single build, since each task now executes
361 an exact source replay and an exact post-transform replay with negative controls. The following
362 secondary analyses diagnose why a task is accepted or rejected and prevent aggregate improvements
363 from masking evaluator faults.

Diagnostic	Required output
Specification grounding	Held-out behavior, interface coverage, copied/paraphrased leakage
Learning dynamics	Fixed token checkpoints, per-seed curves, final-checkpoint interval
Identity and fault injection	Exact IDs, denominator failures, per-family strict-union recall
Retention, cost, decontamination	Cohort funnel, runtime cost, frozen overlap report
Adjudication and grader parity	Blinded weighted error change and per-instance cross-grader agreement
Sensitivity, strata, retrospective audit	Threshold stress, scoped subgroup effects, row-linked transition matrix

Table 7: Secondary analyses and their required outputs.

364 E LLM Usage in the Core Method

365 Language models are core components of environment setup, test-interface organization, and specifi-
366 cation generation. The documentation agent uses repository browsing, test reading, symbol-source
367 retrieval, and runtime-introspection tools. The pipeline records the model version, prompt hash,
368 temperature, and seed per artifact so that construction is reproducible; writing assistance for this
369 manuscript is separate from the scientific pipeline.

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